SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning ***(BL 5)***

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Industry Problem Solving

Code of Conduct:

*I **P.Sai Ganesh Reddy(192525111)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P.Sai Ganesh Reddy

Annexure A

Short Answer Test

1, A retail company wants to recommend products to customers based on past purchases. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity is measured, how recommendations are generated, and how the system adapts to new customer data.

2. A manufacturing industry needs to predict machine output based on sensor readings that vary over time. Propose a solution using Locally Weighted Regression. Explain how local models improve prediction accuracy and how the system handles noisy or incomplete data.

3. A healthcare company aims to classify patients into risk categories using historical medical data. Suggest a model using KNN or RBF networks. Explain how the algorithm handles large datasets, feature selection, and accuracy improvement in real-world scenarios.

4. An e-commerce platform wants to detect similar products for dynamic pricing strategies. Design a solution using Case-Based Learning. Discuss how similarity metrics are defined, how past cases are reused, and how the system scales with increasing product data.

5. A banking system wants to detect potential loan defaulters based on customer financial history and transaction patterns. Design a solution using **K-Nearest Neighbour or Logistic Regression**. Explain how similarity or decision boundaries are used, how features are selected, and how the system adapts to new customer data over time.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. A retail company wants to recommend products to customers based on past purchases. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity is measured, how recommendations are generated, and how the system adapts to new customer data.

Answer

A retail company can use **K-Nearest Neighbour (KNN)** to recommend products based on customers' previous purchases. Each customer is represented using features such as product categories purchased, purchase frequency, spending amount, and ratings.

- **Similarity Measurement:**

  Similarity can be measured using Euclidean distance or cosine similarity. Customers with similar purchasing behaviour are considered neighbours.

- **Recommendation Generation:**

  For a new customer, the system identifies the K most similar customers. Products purchased or highly rated by these neighbours but not yet purchased by the new customer are recommended. **Adaptation to New Data:**

  When a customer makes a new purchase, the purchase history is added to the dataset. The nearest neighbours are recalculated when a new recommendation is required.

- **Conclusion:**

  KNN provides a simple recommendation system that can adapt to changing customer preferences without requiring complete model retraining.

2. A manufacturing industry needs to predict machine output based on sensor readings that vary over time. Propose a solution using Locally Weighted Regression. Explain how local models improve prediction accuracy and how the system handles noisy or incomplete data.

Answer

A manufacturing industry can use **Locally Weighted Regression (LWR)** to predict machine output from sensor readings such as temperature, pressure, vibration, speed, and operating time.

In LWR, greater weights are assigned to training observations that are closer to the current input. A common weight function is:

$$\mathbf{w_i = \exp(-||x - x_i||^2 / 2\tau^2)}$$

where $\boldsymbol{\tau}$ controls the size of the local neighbourhood.

For a new sensor reading, nearby historical observations receive higher weights and are used to build a local regression model. This allows the system to capture changing machine behaviour more effectively than a single global regression model.

Noisy data can be handled using suitable preprocessing, filtering, normalization, and appropriate bandwidth selection. Missing sensor values can be handled using imputation or by excluding unreliable observations.

Conclusion:

Locally Weighted Regression can improve prediction accuracy when machine behaviour changes over time because it focuses mainly on observations similar to the current operating condition.

3. A healthcare company aims to classify patients into risk categories using historical medical data. Suggest a model using KNN or RBF networks. Explain how the algorithm handles large datasets, feature selection, and accuracy improvement in real-world scenarios.

Answer

A healthcare company can classify patients into **low-risk, medium-risk, and high-risk** categories using KNN or an RBF network. Medical features such as age, blood pressure, glucose level, cholesterol, BMI, and previous medical history can be used.

- **Using KNN**

  KNN calculates the distance between a new patient's data and historical patient records. The K nearest patients are selected, and the most common risk category among them is assigned to the new patient.

- **Using RBF Network**

  An RBF network uses radial basis functions to measure the similarity between the input patient data and learned centres. The resulting values are passed to an output layer for classification.

- **Handling Large Datasets**

  For large datasets, efficient indexing, dimensionality reduction, sampling, or approximate nearest-neighbour methods can be used.

- **Feature Selection**

  Irrelevant or redundant medical features should be removed. Important features can be selected using statistical methods or feature-importance techniques.

- **Accuracy Improvement**

  Accuracy can be improved through normalization, selecting an appropriate value of K, cross-validation, feature selection, and balancing the classes.

- **Conclusion:**
  KNN and RBF networks can classify patients effectively when relevant medical features are properly selected and the data is appropriately preprocessed.

4. An e-commerce platform wants to detect similar products for dynamic pricing strategies. Design a solution using Case-Based Learning. Discuss how similarity metrics are defined, how past cases are reused, and how the system scales with increasing product data.

Answer

An e-commerce platform can use **Case-Based Learning (CBL)** to detect similar products and support dynamic pricing. Each previous product case can contain product category, brand, specifications, demand, historical price, sales volume, and customer ratings.

- **Similarity Metrics**

  Similarity can be calculated using numerical distance measures such as Euclidean distance and categorical matching. Features can be normalized so that one feature does not dominate the similarity calculation.

- **Reusing Past Cases**

  When a new product is introduced, the system searches its database for similar historical products. The prices and market behaviour of these cases are used to estimate an appropriate price for the new product.

- **Scaling**

  As product data increases, efficient indexing, clustering, dimensionality reduction, and retrieval techniques can be used to find similar cases quickly.

  The system can also update its case database whenever new sales and pricing information becomes available.

- **Conclusion:**
  Case-Based Learning allows an e-commerce platform to reuse historical pricing experiences and adapt pricing decisions as new product and market data becomes available.

5. A banking system wants to detect potential loan defaulters based on customer financial history and transaction patterns. Design a solution using **K-Nearest Neighbour or Logistic Regression.** Explain how similarity or decision boundaries are used, how features are selected, and how the system adapts to new customer data over time.

Answer

A banking system can predict potential loan defaulters using **KNN or Logistic Regression**. Customer features can include income, credit score, loan amount, debt-to-income ratio, transaction history, repayment history, and account balance.

- **Using KNN**

  KNN measures the similarity between a new customer and historical customers. If most of the nearest customers are defaulters, the new customer can be classified as a potential defaulter.

- **Using Logistic Regression**

  Logistic Regression calculates the probability of default:

$$\mathbf{P(Default) = 1 / (1 + e^{-z})}$$

  where **z** is a weighted combination of the selected financial features.

- **Feature Selection**

  Features that strongly influence loan repayment should be selected, while irrelevant and highly redundant features should be removed. Feature scaling is particularly important for KNN.

- **Adapting to New Data**

  New repayment outcomes can be added to the training dataset. The KNN system can immediately use these new cases, while a Logistic Regression model can periodically be retrained with updated customer data.

- **Conclusion:**

  KNN is useful for similarity-based prediction, while Logistic Regression provides an interpretable probability of default. Both approaches can be updated using new customer and repayment information.